

Solutions to JEE Main - 3 | JEE-2022

PHYSICS

SECTION-1

- 1.(C) The particle moves in a circle in the horizontal plane, so its acceleration is directed horizontally. Therefore, the vertical component of tension must balance the weight of the particle.

$$\text{So, } T \cos \theta = mg$$

$$\Rightarrow T = \frac{mg}{\cos \theta}$$

- 2.(B) Work, $W = \vec{F} \cdot \vec{s} = (10\hat{i}) \cdot (\hat{i} - \hat{j}) = 10 \text{ J}$

- 3.(C) We know that the maximum height becomes e^2 times after each rebound

$$\text{Therefore, } e^2 = \frac{h_1}{h_0}$$

$$\text{So, after the second rebound, the maximum height, } h_2 = e^2 h_1 = \frac{h_1^2}{h_0}$$

- 4.(D) We know that the force of friction, $f = \mu N = \mu mg \cos \theta$

$$\text{Therefore, the net force down the inclined plane, } F = mg \sin \theta - \mu mg \cos \theta$$

$$\text{Therefore, the acceleration of the block, } a = \frac{F}{m} = g(\sin \theta - \mu \cos \theta) = (10)(0.6 - (0.5)(0.8)) = 2 \text{ m/s}^2$$

Now, using $v^2 = 2as$, the velocity of the block after it has moved a distance 1 m,

$$v = \sqrt{2(2)(1)} = 2 \text{ m/s}$$

- 5.(A) Taking the origin at the location of A,

$$X_{CM} = \frac{m(0) + 2m(L) + 4m(2L)}{m + 2m + 4m} = \frac{10L}{7}$$

$$\text{Therefore, Distance of CM from A} = \frac{10L}{7}$$

$$\text{Distance of CM from B} = \frac{10L}{7} - L = \frac{3L}{7}$$

- 6.(B) Let the tension in the string be T

Let the coefficient of friction between the block of mass m_1 and the table be μ

Since the blocks move with constant velocity, the net force on both of them must be zero

Also, it is clear that the blocks must move such that the block of mass m_2 is going down

$$\text{Therefore, } T = m_2 g \quad \text{and} \quad T = \mu m_1 g$$

$$\text{Solving, we get } \mu = \frac{m_2}{m_1}$$

- 7.(A) Let us assume that the block strikes the spring once and then stops at A.

Then, conserving energy between the initial and final instants,

$$\text{Loss in kinetic energy} = \text{Heat loss in friction}$$

$$\Rightarrow \frac{1}{2} mu^2 = \mu mg(2L) \Rightarrow u_{\min} = \sqrt{4\mu gL}$$

If the initial velocity of the block is greater than $u_{\min}$, it will not stop at A and move further left.

- 8.(D)** Let the length of the string be L and the velocity of the particle when it is at the lowest point be v

Conserving energy, $mgL(1 - \cos\theta) = \frac{1}{2}mv^2 \Rightarrow v^2 = 2gL(1 - \cos\theta)$

Therefore, acceleration at the lowest point, $a = \frac{v^2}{L} = 2g(1 - \cos\theta) = 2g\left(1 - \frac{1}{\sqrt{2}}\right) = (2 - \sqrt{2})g$

- 9.(D)** Let the displacement of the plank be x

Then, the displacement of the man relative to the ground is $L + x$

Since the external force on the system is zero, and the CM was initially at rest, the displacement of the CM must be zero

Therefore, $X_{CM} = \frac{M(L + x) + m(x)}{M + m} = 0$

$$\Rightarrow x = -\left(\frac{M}{M + m}\right)L$$

- 10.(D)** After the block has moved downwards by a distance x , the force on it, $F = kx - mg$

Therefore, its acceleration, $a = \frac{F}{m} = \left(\frac{k}{m}\right)x - g$

We can see that the graph must be a straight line with positive slope, and the value of a at $x = 0$ is $-g$

- 11.(A)** Let us assume that the acceleration of the system is a upwards

Let the force exerted by the rope on the block be T

Then, $F - T - mg = ma$ and $T - Mg = Ma$

Adding the equations, we get $a = \frac{F}{M + m} - g$

Therefore, $T = Mg + Ma = \left(\frac{M}{M + m}\right)F$

- 12.(C)** Angular velocity of the disk, $\omega = \alpha t$

Block A slips when $\omega^2 r_1 = \mu_1 g \Rightarrow t_1 = \sqrt{\frac{\mu_1 g}{r_1}} \frac{1}{\alpha}$

Similarly, block B slips when $t_2 = \sqrt{\frac{\mu_2 g}{r_2}} \frac{1}{\alpha}$

So, block A slips before block B does if

$$t_1 < t_2 \Rightarrow \mu_1 r_2 < \mu_2 r_1$$

- 13.(C)** Since the net external force on the system of the disks is always zero (their weight is balanced by the vertically acting normal reactions from the table), the centre of mass moves with constant velocity.

So, $\vec{v}_{CM} = \frac{m(4\hat{i}) + 2m(\hat{i} - 3\hat{j})}{m + 2m} = 2\hat{i} - 2\hat{j}$

Since at $t = 0$, the disks are both at the origin, the CM is also at the origin

Therefore, at $t = 2$ s, the position of the CM, $\vec{r}_{CM} = (0\hat{i} + 0\hat{j}) + \vec{v}_{CM}(2) = 4\hat{i} - 4\hat{j}$

- 14.(A)** Net force on the particle, $\vec{F} = \frac{d}{dt}(\vec{p}(t)) = -A\omega\sin(\omega t)\hat{i} + B\omega\cos(\omega t)\hat{j}$

- 15.(A) Because the particle strikes the wall horizontally, the time taken to go from the ground to the wall, and the time taken to go from the wall to the ground again is equal

This time is $t = \frac{u \sin \theta}{g}$

Let the horizontal component of velocity after the collision with the wall be v_x

Then, $L = v_x t = v_x \left(\frac{u \sin \theta}{g} \right) \Rightarrow v_x = \frac{gL}{u \sin \theta}$

We know that before hitting the ground, the vertical component of velocity, $v_y = u \sin \theta$

So, the angle with the horizontal made by the velocity of the particle just before it strikes the ground,

$$\phi = \tan^{-1} \left(\frac{v_y}{v_x} \right) = \tan^{-1} \left(\frac{u^2 \sin^2 \theta}{gL} \right)$$

- 16.(B) Let the initial direction of motion of the particle be the +X direction

Then, we can assume that the given force, $\vec{F} = 15(\cos \theta \hat{i} + \sin \theta \hat{j}) = 15(0.6 \hat{i} + 0.8 \hat{j}) = 9 \hat{i} + 12 \hat{j}$

Since this force is constant, the particle's acceleration is also constant. The components of this acceleration are:

$$a_x = \frac{9}{1} = 9 \text{ m/s}^2 \quad \text{and} \quad a_y = \frac{12}{1} = 12 \text{ m/s}^2$$

Therefore, after 5 seconds, the components of velocity of the particle are:

$$v_x = 35 + 9(5) = 80 \text{ m/s} \quad \text{and} \quad v_y = 12(5) = 60 \text{ m/s}$$

So, the velocity of the particle, $v = \sqrt{v_x^2 + v_y^2} = 100 \text{ m/s}$

- 17.(A) Let the acceleration of the blocks B and C be a_1 and a_2 respectively (both downwards) and let the acceleration of block A be a towards right

Let the tension in the string connected to blocks B and C be T

Then, we know that the tension in the string connected to block A is $2T$ and the accelerations of the blocks are related as:

$$2a = a_1 + a_2$$

From the force diagrams,

Block A: $2T = 4a$

Block B: $1g - T = 1a_1$

Block C: $3g - T = 3a_2$

Solving, we get $T = \frac{6g}{7}$, $a_1 = \frac{g}{7}$, $a_2 = \frac{5g}{7}$, $a = \frac{3g}{7}$

- 18.(B) Let the forces be directed along the +X, +Y and +Z directions

Then, the acceleration of the particle, $\vec{a} = \left(\frac{F}{m} \right) (\hat{i} + \hat{j} + \hat{k})$

Therefore, $a = \frac{\sqrt{3}F}{m}$

So, the velocity of the particle at $t = t_0$, $v = at_0 = \frac{\sqrt{3}F t_0}{m}$

Therefore, its kinetic energy, $K = \frac{1}{2}mv^2 = \frac{3F^2 t_0^2}{2m}$

Because of symmetry, the work done by all the forces is equal. Hence, the work done by any one

$$\text{force, } W_1 = \frac{K}{3} = \frac{F^2 t_0^2}{2m}$$

Alternative solution

We already know the acceleration of the particle, $\vec{a} = \left(\frac{F}{m}\right)(\hat{i} + \hat{j} + \hat{k})$

Since the acceleration is constant, the displacement of the particle until $t = t_0$,

$$\vec{s} = \frac{1}{2} \vec{a} t_0^2 = \left(\frac{F t_0^2}{2m}\right)(\hat{i} + \hat{j} + \hat{k})$$

Therefore, the work done by one of the forces, say $\vec{F}_1 = F \hat{i}$,

$$W_1 = \vec{F}_1 \cdot \vec{s} = \frac{F^2 t_0^2}{2m}$$

19.(A) We are given that $U(x) = 16 + \frac{1}{4}(x-2)^2$

Therefore, $\frac{dU}{dx} = \frac{1}{2}(x-2)$

And, $\frac{d^2U}{dx^2} = \frac{1}{2}$

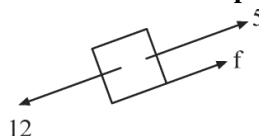
We can see that at $x = 2$, the first derivative of U is zero and the second derivative of U is positive.

Therefore, the position $x = 2$ is a stable equilibrium.

20.(C) Component of gravity down the plane $= mg \sin \theta = (2)(10)(0.6) = 12 \text{ N}$

Spring force, acting up the plane $= kx = (50)(0.1) = 5 \text{ N}$

Since the resultant of the two forces above is **7 N down the plane**, friction acts up the plane.



Maximum possible value of static friction $= \mu mg \cos \theta = (0.2)(2)(10)(0.8) = 3.2 \text{ N}$

Since this maximum value of friction is less than 7 N, the block slips down the plane, and its acceleration is:

$$a = \frac{7 - 3.2}{2} = 1.9 \text{ m/s}^2 \quad (\text{down the plane})$$

SECTION-2

1.(36) Acceleration of the particle, $a = \frac{F}{m} = 3t$

Therefore, the velocity of the particle, $v(t) = \int_0^t a dt = \int_0^t (3t) dt = \frac{3}{2} t^2$

So, the instantaneous power being delivered to the particle, $P(t) = Fv = (3t) \left(\frac{3}{2} t^2\right) = \frac{9}{2} t^3$

At $t = 2 \text{ s}$, $P = 36 \text{ W}$

- 2.(2) Let the velocity of the block at the top-most point of the track be v

Then, conserving energy:

Loss in kinetic energy = Gain in potential energy

$$\frac{1}{2}m(u^2 - v^2) = mg(0.8) \Rightarrow v = 4 \text{ m/s}$$

After leaving the track, the motion of the block is like a projectile. So, its velocity at the highest point

in its path is equal to the horizontal component of its velocity, i.e. $v \cos 60^\circ = (4)\left(\frac{1}{2}\right) = 2 \text{ m/s}$

- 3.(15) Conserving momentum: $1(6) + 3(-8) = 1v_1 + 3v_2$

Newton's experimental law: $v_2 - v_1 = 6 + 8$

Solving, we get $v_1 = -15 \text{ m/s}$ and $v_2 = -1 \text{ m/s}$

- 4.(4) Let the speed of the ball after the collision be v

Also, since after the collision, the ball travels in a direction perpendicular to its initial direction, the

ball's velocity now makes angle 30° with the horizontal

Since the component of the velocity of the ball parallel to the ground remains unchanged during the collision,

$$u \cos 60^\circ = v \cos 30^\circ \Rightarrow v = \frac{u}{\sqrt{3}} = \frac{10}{\sqrt{3}} \text{ m/s}$$

$$\text{Therefore, loss in kinetic energy} = \frac{1}{2}mu^2 - \frac{1}{2}mv^2 = \frac{1}{2}(0.12)\left(10^2 - \left(\frac{10}{\sqrt{3}}\right)^2\right) = 4 \text{ J}$$

- 5.(2) Let the acceleration of the blocks be a and the tension in the string be T

Then, $m_1g - T = m_1a$ and $T - m_2g = m_2a$

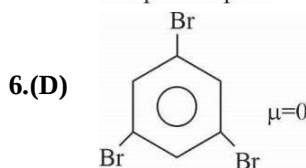
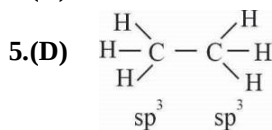
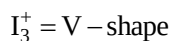
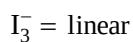
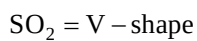
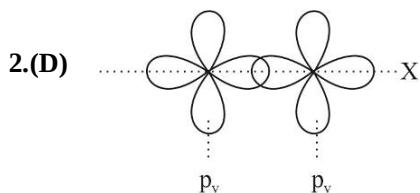
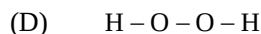
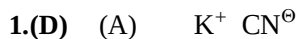
$$\text{Adding the equations, we get } a = \left(\frac{m_1 - m_2}{m_1 + m_2}\right)g = \left(\frac{0.6 - 0.4}{0.6 + 0.4}\right)(10) = 2 \text{ m/s}^2$$

Displacement of any one block during 1 second, $s = \frac{1}{2}(2)(1)^2 = 1 \text{ m}$

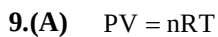
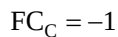
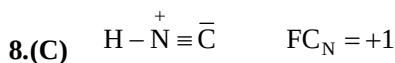
Therefore, the height difference after 1 second, $h = 2s = 2 \text{ m}$

Chemistry

SECTION-1



More electronegativity more partial charge, more strength of H-bond



$$PV = \frac{\text{mass}}{M_0} \cdot RT$$

$$1 \times M_0 = d \times R \times T$$

$$1 \times 44 = d \times 0.082 \times 546$$

$$d = 0.98 \text{ g/L}$$

$$10.(C) \quad \lambda = \frac{h}{mv} \quad \text{and} \quad V = \sqrt{\frac{3RT}{M}}$$

$$\frac{\lambda_1}{\lambda_2} = \sqrt{\frac{T_2}{T_1}}$$

$$\frac{\lambda_1}{\lambda_2} = \sqrt{\frac{600}{300}} = \sqrt{\frac{2}{1}}$$

$$11.(B) \quad PV = nRT$$

$$V = \frac{nR}{P} \cdot T$$

$$\log V = \log \frac{nR}{P} + \log T$$

$$y = C + x$$

$$C = 0$$

$$\log \frac{nR}{P} = 0$$

$$\frac{nR}{P} = 1$$

$$\frac{n \times 0.0821}{0.1642} = 1$$

$$n = 2$$

$$12.(A) \quad V_{\text{rms}} = \sqrt{\frac{3RT}{M}}$$

$$\frac{V_{\text{rms}(1)}}{V_{\text{rms}(2)}} = \sqrt{\frac{T_1}{T_2}} \times \frac{M_{\text{av}}}{M_1}$$

$$= \sqrt{\frac{100}{400}} \times \frac{9}{16} = \frac{3}{8}$$

$$\left(M_{\text{av}} = (X_1 M_1 + X_2 M_2) = \frac{1 \times 2 + 1 \times 16}{2} = 9 \text{ gmol}^{-1} \right)$$

$$13.(B) \quad PV = nRT$$

$$n, T, R = \text{constant}$$

$$P_1 V_1 = P_2 V_2$$

$$10 \times 1 = 1 \times V_2$$

$$V_2 = 10 \text{ ml}$$

$$14.(C) \quad PV = nRT$$

$$P = \frac{nR}{V} \cdot T$$

$$y = m \cdot x$$

$$V \uparrow \Rightarrow \frac{nR}{V} \downarrow \Rightarrow m \downarrow \Rightarrow \text{slope} \downarrow$$

$$15.(C) \quad PV = nRT \quad \text{if } n=1$$

$$PV = RT$$

$$V = \frac{RT}{P}$$

16.(A) Total mole = $\left(\frac{w}{2} + \frac{w}{16} + \frac{w}{32}\right) = \frac{19x}{32}$

For CH_4 , $P_{\text{CH}_4} V = n_{\text{CH}_4} RT$

Also, $P_{\text{total}} V = n_{\text{total}} RT$

$$\therefore \frac{P_{\text{CH}_4}}{P_{\text{total}}} = \frac{n_{\text{CH}_4}}{n_{\text{total}}} = \frac{w/16}{\frac{19x}{32}} = \frac{2}{19}$$



acid base

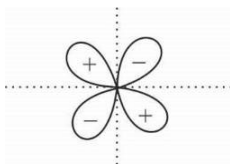
meq = meq

$$m \text{ mol} \times 2 = m \text{ mol} \times 1$$

$$0.3 \times V \times 2 = 0.5 \times 200 \times 1$$

$$V = 166.67 \text{ ml}$$

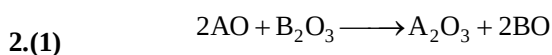
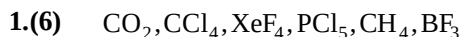
18.(B)



19.(A) % of N = $\frac{\text{mass of Nitrogen}}{\text{total mass}} \times 100 = \frac{28}{80} \times 100 = 35\%$

20.(A) $sp^3d^2 = s + p_x + p_y + p_z + d_{z^2} + d_{x^2-y^2}$

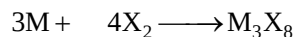
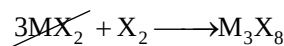
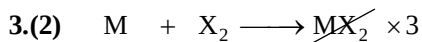
SECTION-2



$$t = 0 \quad 10 \quad 2 \quad - \quad -$$

$$\text{LR} = \text{B}_2\text{O}_3$$

$$\text{Mole of BO} = 4 \times \frac{25}{100} = 1 \text{ mole}$$



$$t = 0 \quad 10 \quad 8 \quad -$$

$$\text{M}_3\text{X}_8 = 2 \text{ mole}$$

4.(2) Mole of ions = $\frac{38}{95} \times 3 + \frac{23.4}{58.5} \times 2 = 2 \text{ mole}$



Mathematics

SECTION-1

- 1.(B) V I R A T
 Alphabetical order
 "A I R T V"
 $A_{-----} = 24$
 $I_{-----} = 24$
 $R_{-----} = 24$
 $T_{-----} = 24$
 $VA_{-----} = 6$
 $VI A_{--} = 2$
 $V I R A T - 105^{th}$
- 2.(A) $\sin \alpha = -\frac{1}{\sqrt{2}} \Rightarrow \alpha = \pi + \frac{\pi}{4}, 2\pi - \frac{\pi}{4}$
 $\cos \beta = -\frac{1}{2} \Rightarrow \beta = \pi - \frac{\pi}{3}, \pi + \frac{\pi}{3}$
 $\tan \gamma = -\sqrt{3} \Rightarrow \gamma = \pi - \frac{\pi}{3}, 2\pi - \frac{\pi}{3}$ $\therefore$ Sum of all values $= 8\pi - \frac{2\pi}{3} = \frac{22\pi}{3}$
- 3.(D) P K AA RR
 $2! \times {}^3C_2 \times 2! = 12$
- 4.(B) $25 \begin{cases} B_1, B_2, \dots, B_7 \\ \begin{cases} \begin{cases} \begin{cases} 9 \\ 6 \\ 3 \end{cases} \end{cases} \end{cases} \end{cases}$ $10! \times 2 \times 2$ (increasing or decreasing)
- 5.(C) $7^8 - 7$
- 6.(D) $\cos\left(\frac{\pi}{4}\right)\cos\left(\frac{\pi}{8}\right)\cos\left(\frac{\pi}{16}\right)\dots\cos\left(\frac{\pi}{2^n}\right)$ $\left(A = \frac{\pi}{2^n}\right)$
 $\cos A \cos 2A \dots \cos(2^{n-2}A) = \frac{1}{2^{n-1}} \frac{\sin(2^{n-1}A)}{\sin A}$
- 7.(C) $B_1 + B_2 + B_3 + B_4 = 11$
 ${}^{14}C_3 = \frac{14 \times 13 \times 12}{6} = 364$
- 8.(A) $x_1 x_2 x_3 = 150 = 2 \times 3 \times 5^2$
 Number of integral solution $= {}^3C_1 \times {}^3C_1 \times {}^4C_2 = 9 \times 6 = 54$
- 9.(C) ${}^{18}C_4 - {}^9C_4 \times 2^4$
 $\frac{18 \times 17 \times 16 \times 15}{24} - \frac{9 \times 8 \times 7 \times 6}{24} \times 2$
 $3060 - 2016 = 1044$
- 10.(D) ${}^6C_3 \times 5 \times 5 \times 5 \times 21 \times 21 \times 21 = 20 \times (105)^3$
- 11.(A) $x_1 + x_2 + x_3 + x_4 + x_5 = 7$
 $\geq 0 \quad \geq 1 \quad \geq 1 \quad \geq 1 \quad \geq 0 = 4$
 x_1 is number of person before 1st selected person 4
 x_2 is number of person b/w first and second selected person.

$$12.(A) \quad S = -2\sin\left(\frac{1}{2}\right)\cos\left(\frac{5}{2}\right)$$

$$S = -2\sin\left(\frac{1}{2}\right)\cos\left(\frac{5}{2}\right)$$

+ve -ve

$$S > 0$$

$$C = 2\sin\left(\frac{1}{2}\right)\sin\left(\frac{5}{2}\right)$$

+ve +ve

$$C > 0$$

$$13.(B) \quad a = (1 + \sin 2\theta)^2$$

$$b = (1 - \sin 2\theta)^2 \Rightarrow ab = \cos^4 2\theta \leq 1$$

$$14.(D) \quad 2 \tan x = \alpha(1 + \tan^2 x)$$

$$\Rightarrow \frac{2 \tan x}{1 + \tan^2 x} = \alpha \Rightarrow \sin 2x = \alpha$$

$\therefore \angle B$ and $\angle C$ satisfy this equation

$$\Rightarrow \sin 2B = \sin 2C$$

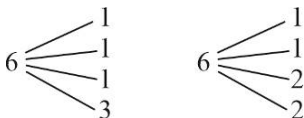
But angle $B > \text{angle } C$

$$\Rightarrow 2C = \pi - 2B \Rightarrow B + C = \frac{\pi}{2} \quad \therefore \angle A = \frac{\pi}{2}$$

$$15.(C) \quad {}^5C_3 \times {}^8C_7 + {}^5C_4 \times {}^8C_6 + {}^5C_5 \times {}^8C_5$$

$$80 + 140 + 56 = 276$$

$$16.(B)$$



$${}^4C_1 \times \frac{6!}{3!} + {}^4C_2 \times \frac{6!}{2!2!} = 13 \times 5! = 1560$$

$$13 \times 5! = 120 \times 13 = 1560$$

$$17.(B) \quad 1 - 2\sin^2 2x \sin^2 2x = \sin 2x \cos 2x$$

$$2t^2 + t - 1 = 0 \quad \frac{\sin 4x}{2} = -1 \text{ or } 1/2$$

$$2t^2 + 2t - t - 1 = 0 \quad \sin 4x = 1$$

$$(2t - 1)(t + 1) = 0$$

$$t = -1 \text{ or } 1/2$$

$$18.(B) \quad {}^{10}C_1 \times {}^9C_1 \times {}^{10}C_2 \times 8! ; \quad 10! {}^{10}C_2$$

$$19.(D) \quad \frac{\sin 3x}{2\cos 2x + 1} = \frac{1}{2}$$

$$\Rightarrow 2(3\sin x - 4\sin^3 x) = 2(1 - 2\sin^2 x) + 1 \Rightarrow 6\sin x - 8\sin^3 x = 2 - 4\sin^2 x + 1$$

$$\Rightarrow 8\sin^3 x - 4\sin^2 x - 6\sin x + 3 = 0 \Rightarrow 4\sin^2 x(2\sin x - 1) - 3(2\sin x - 1) = 0$$

$$\sin x = \frac{1}{2} \text{ or } \sin^2 x = \frac{3}{4} \quad (\text{Not possible as } 2\cos 2x + 1 = 0)$$

$$x = n\pi + (-1)^n \frac{\pi}{6}$$

$$20.(A) \quad {}^{12}C_3 - {}^5C_3 = \frac{12 \times 11 \times 10}{6} - \frac{5 \times 4 \times 3}{4}$$

$$220 - 10 = 210$$

SECTION-2

$$1.(6) \quad 2\sin^2 x + \sin x - 1 = 0$$

$$2\sin^2 x + 2\sin x - \sin x - 1 = 0$$

$$2\sin x(\sin x + 1) - 1(\sin x + 1) = 0$$

$$\sin x = -1 \quad \text{or} \quad \sin x = \frac{1}{2}$$

$$[0, 4\pi] \quad [0, 4\pi]$$

$$2 \text{ solution} \quad \text{or} \quad 4 \text{ solution}$$

$$2.(6) \quad \sin^2 \theta = t$$

$$\frac{t^2}{5} + (1-t)^2 = \frac{1}{6}$$

$$(6t-5)^2 = 0$$

$$t = \frac{5}{6}$$

$$\sin^2 \theta = \frac{5}{6}; \quad \cos^2 \theta = \frac{1}{6} \Rightarrow \sec^2 \theta = 6$$

$$3.(27) \quad \text{AAAA HH M BRT}$$

$$4A-1$$

$$3A+1D - {}^1C_1 \times {}^5C_1 \times {}^5C_1 \times \frac{4!}{3!}$$

$$2A+20A - {}^2C_2 \times \frac{4!}{2!2!}$$

$$2A+20A - {}^2C_2 \times \frac{4!}{2!2!}$$

$$2A+2D \rightarrow {}^2C_1 \times {}^5C_2 \times \frac{4!}{2!}$$

$$\text{All } D \rightarrow {}^6C_4 \times 4!$$

$$\text{Total} = 627$$

$$4.(378) \quad 2^5 \cdot 3^3 \cdot 5^5 \cdot 2^3 \cdot 3^3$$

$$2^8 \cdot 3^6 \cdot 5^5$$

$$9 \times 7 \times 6 = 378$$

$$5.(300) \quad {}^6C_2 \times {}^5C_2 + {}^5C_2 \times {}^4C_2 + {}^6C_2 \times {}^4C_2$$

$$15 \times 10 + 10 \times 6 + 15 \times 6 = 150 + 60 + 90 = 300$$